Nancy

AI/ML Engineer | NLP, RAG & Vector Search Systems
Sonipat, Haryana, India | nnydv498@gmail.com | +91-8168750597 | linkedin.com/in/nancyy56 | github.com/Nancyy56

Summary

AI/ML engineer with production experience building RAG pipelines, semantic search systems, and LLM-integrated backend services using Python, NLP, transformer models, and vector databases. Currently shipping AI-assisted flight-operations software in a live product team.

Experience

International Flight Operations Academy (IFOA) India LLP New Delhi
AI Integration Developer Intern *Aug 2026 – Present*

- Build AI-assisted flight-dispatch features on a React/TypeScript and Node.js platform with a distributed India–Switzerland engineering team.
- Engineered a route weather-corridor service sampling 25 forecast frames in parallel under a concurrency gate, filtering weather per pixel against the great-circle flight path.
- Replaced leaking in-memory caches with a persistent SQLite TTL cache, cutting repeat route queries to sub-second reads.
- Integrated LLM parsing (GPT-4o / Gemini) into NOTAM, ATIS, and MEL/CDL chat responses, and built vertical-profile and 3D route visualizations of turbulence, icing, and terrain data.

Projects

GovLens – Government Document RAG & Citation System | *Python, FastAPI, Qdrant, PostgreSQL, Ollama* GitHub

- Built a modular RAG pipeline for government notifications with rule-based metadata extraction, classification, and page-preserving chunking for citation-accurate retrieval.
- Engineered a dense retrieval layer using **BAAI/bge-small-en-v1.5** embeddings with Qdrant cosine similarity search via a FastAPI service backed by PostgreSQL.
- Designed a deterministic citation engine and supersession tracker grounding LLM answers (Ollama qwen3:8b) in verifiable source text, reducing hallucinated responses.

BizIntel AI – AI-Powered Business Intelligence System | *Python, FastAPI, RAG, Prophet, ChromaDB* GitHub

- Engineered an end-to-end AI BI system integrating RAG, query routing, anomaly detection, and time-series forecasting for grounded, hallucination-mitigated insights.
- Reduced forecasting error by **∼48%** over a moving average baseline using Prophet with MAE/RMSE-driven evaluation and backtesting.
- Implemented a hybrid query router (rules, ML classification, LLM intent resolution) with Z-score and Isolation Forest anomaly detection on business KPIs.

DevScope – AI-Powered Codebase Intelligence System | *Python, FastAPI, SentenceTransformers, ChromaDB, RAG* GitHub

- Designed a modular ingestion pipeline parsing Python repositories with AST, extracting functions and classes as structured units for semantic indexing.
- Implemented semantic code search retrieving snippets in under 300ms using SentenceTransformer embeddings in a persistent ChromaDB vector database.
- Built a RAG architecture combining retrieval, context construction, and prompt generation, deployed as a FastAPI REST service and CLI for natural-language codebase queries.

Education

Deenbandhu Chhotu Ram University of Science and Technology (DCRUST) Sonipat, Haryana
Bachelor of Technology – Computer Science and Engineering *Sep 2023 – Jul 2027*
CGPA: 8.32/10 Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks

Technical Skills

Languages: Python, C++, SQL, JavaScript, TypeScript
AI & Machine Learning: Supervised & Unsupervised Learning, Scikit-learn, XGBoost, PyTorch, Large Language Models (LLMs)
NLP & Semantic Search: Sentence Transformers (SBERT), BERT, TF-IDF, Tokenization, Dense Retrieval, spaCy, NLTK
Vector DBs & RAG: ChromaDB, Qdrant, ANN Search, Retrieval-Augmented Generation (RAG), Vector Embeddings
MLOps & Backend: FastAPI, Streamlit, Apache Airflow, PostgreSQL, SQLite, REST APIs, Model Serving, Node.js
Data & Tools: Pandas, NumPy, Matplotlib, Prophet, MySQL, Git, GitHub, Google Colab, GCP

Achievements & Activities

Innovation Award – Unisys Hackathon 2024. **150+ DSA problems** solved on LeetCode (C++).
ML Lead, GDG DCRUST (2024–25) – Organized AI/ML contests and technical sessions for 100+ students.
Machine Learning Certification – *Supervised Machine Learning*, Stanford Online & DeepLearning.AI (Coursera).